DBMS LAB 7 — PL/SQL BASICS

8 Questions with Oracle PL/SQL Answers

Note: Q5 uses BONUS, DEPT_BUDGET and PRJ_FUND as introduced in the supplied manual's earlier ALTER TABLE exercises. Ensure those columns exist before running Q5.

1. Print first 5 department numbers.

```
SET SERVEROUTPUT ON;
BEGIN
  FOR i IN 1..5 LOOP
    DBMS_OUTPUT.PUT_LINE('Department Number: ' || i);
  END LOOP;
END;
/
```

2. Accept employee number and display employee name, department and salary in the required format.

```
SET SERVEROUTPUT ON;
DECLARE
  v_empno EMP.EMPNO%TYPE := &empno;
  v_ename EMP.ENAME%TYPE;
  v_dname DEPT.DNAME%TYPE;
  v_sal   EMP.SAL%TYPE;
BEGIN
  SELECT E.ENAME, D.DNAME, E.SAL
  INTO v_ename, v_dname, v_sal
  FROM EMP E JOIN DEPT D ON E.DEPTNO=D.DNO
  WHERE E.EMPNO=v_empno;

  DBMS_OUTPUT.PUT_LINE(v_ename || ' works in ' || v_dname ||
    ' department and draws ' || v_sal || '/- as salary');
END;
/
```

3. Create EMPSAL and calculate HRA, DA, Gross Salary, PF and Net Salary, then insert the result.

```
CREATE TABLE EMPSAL (
  EMPNO NUMBER(10), EMPNAME VARCHAR2(30), SAL NUMBER(10,2),
  HRA NUMBER(10,2), DA NUMBER(10,2), GROSS_SAL NUMBER(10,2),
  PF NUMBER(10,2), NET_SAL NUMBER(10,2)
);

SET SERVEROUTPUT ON;
DECLARE
  v_empno EMP.EMPNO%TYPE := &empno;
  v_name EMP.ENAME%TYPE;
  v_sal EMP.SAL%TYPE;
  v_hra NUMBER(10,2); v_da NUMBER(10,2);
  v_gross NUMBER(10,2); v_pf NUMBER(10,2); v_net NUMBER(10,2);
BEGIN
  SELECT ENAME,SAL INTO v_name,v_sal FROM EMP WHERE EMPNO=v_empno;
  v_hra:=v_sal*0.50; v_da:=v_sal*0.20; v_pf:=v_sal*0.12;
  v_gross:=v_sal+v_hra+v_da; v_net:=v_gross-v_pf;
  INSERT INTO EMPSAL VALUES
    (v_empno,v_name,v_sal,v_hra,v_da,v_gross,v_pf,v_net);
  DBMS_OUTPUT.PUT_LINE('Salary details inserted for '||v_name);
END;
/
```

4. Use a cursor to display the department-wise employee list.

```
SET SERVEROUTPUT ON;
DECLARE
  CURSOR c_emp IS
    SELECT D.DNAME,E.EMPNO,E.ENAME,E.JOB
    FROM DEPT D JOIN EMP E ON D.DNO=E.DEPTNO
    ORDER BY D.DNAME,E.EMPNO;
BEGIN
  FOR r IN c_emp LOOP
    DBMS_OUTPUT.PUT_LINE(r.DNAME || ' -> ' || r.EMPNO || ' ' ||
      r.ENAME || ' (' || r.JOB || ')');
  END LOOP;
END;
/
```

5. Allocate incentives from Prj_Fund according to Prj_Credits; share 70% among employees and add 30% to Dept_Budget.

```
SET SERVEROUTPUT ON;
DECLARE
  CURSOR c_proj IS
    SELECT DNO,PRJ_NO,PRJ_CREDIT,PRJ_FUND FROM PROJ;
  v_fund NUMBER(12,2);
  v_emp_share NUMBER(12,2);
  v_dept_share NUMBER(12,2);
  v_count NUMBER;
BEGIN
  FOR p IN c_proj LOOP
    v_fund:=p.PRJ_FUND*(p.PRJ_CREDIT*0.10);

    SELECT COUNT(*) INTO v_count
    FROM EMP
    WHERE DEPTNO=p.DNO AND PRJ_ID=p.PRJ_NO;

    v_emp_share:=v_fund*0.70;
    IF v_count>0 THEN
      v_emp_share:=v_emp_share/v_count;
      UPDATE EMP
      SET BONUS=NVL(BONUS,0)+v_emp_share
      WHERE DEPTNO=p.DNO AND PRJ_ID=p.PRJ_NO;
    END IF;

    v_dept_share:=v_fund*0.30;
    UPDATE DEPT
    SET DEPT_BUDGET=NVL(DEPT_BUDGET,0)+v_dept_share
    WHERE DNO=p.DNO;
  END LOOP;
  COMMIT;
END;
/
```

6. Use a cursor FOR LOOP to list all projects and their associated tasks.

```
SET SERVEROUTPUT ON;
DECLARE
  CURSOR c_pt IS
    SELECT P.PRJ_NO,P.DNO,P.PRJ_NAME,T.TASK_ID,T.TASK_NAME,T.STATUS
    FROM PROJ P
    LEFT JOIN TASK T ON P.PRJ_NO=T.PRJ_ID AND P.DNO=T.DNO
    ORDER BY P.DNO,P.PRJ_NO,T.TASK_ID;
BEGIN
  FOR r IN c_pt LOOP
    DBMS_OUTPUT.PUT_LINE('Project: '||r.PRJ_NAME||
      ' | Task ID: '||NVL(TO_CHAR(r.TASK_ID),'No Task')||
      ' | Task: '||NVL(r.TASK_NAME,'No Task')||
      ' | Status: '||NVL(r.STATUS,'N/A'));
  END LOOP;
END;
/
```

7. Use a parameterized cursor to display employees for user-given job and department number.

```
SET SERVEROUTPUT ON;
DECLARE
  v_job EMP.JOB%TYPE:='&job';
  v_deptno EMP.DEPTNO%TYPE:=&deptno;

  CURSOR c_emp(p_job EMP.JOB%TYPE,p_deptno EMP.DEPTNO%TYPE) IS
    SELECT EMPNO,ENAME,JOB,SAL FROM EMP
    WHERE UPPER(JOB)=UPPER(p_job) AND DEPTNO=p_deptno;
BEGIN
  FOR r IN c_emp(v_job,v_deptno) LOOP
    DBMS_OUTPUT.PUT_LINE(r.EMPNO||' '||r.ENAME||
      ' | '||r.JOB||' | Salary: '||r.SAL);
  END LOOP;
END;
/
```

8. Use a parameterized cursor to display the first two highest-paid employees on a user-given project.

```
SET SERVEROUTPUT ON;
DECLARE
  v_prj_id EMP.PRJ_ID%TYPE:='&prj_id';

  CURSOR c_top(p_prj_id EMP.PRJ_ID%TYPE) IS
    SELECT ENAME,SAL,DNAME
```

```
    FROM (
      SELECT E.ENAME,E.SAL,D.DNAME,
             ROW_NUMBER() OVER(ORDER BY E.SAL DESC) RN
      FROM EMP E JOIN DEPT D ON E.DEPTNO=D.DNO
      WHERE E.PRJ_ID=p_prj_id
    )
    WHERE RN<=2
    ORDER BY SAL DESC;
BEGIN
  FOR r IN c_top(v_prj_id) LOOP
    DBMS_OUTPUT.PUT_LINE(r.ENAME||' | Salary: '||r.SAL||
      ' | Department: '||r.DNAME);
  END LOOP;
END;
/
```